An Empirical Comparison of Policy Sharing Strategies in Cooperative Multi-Agent Reinforcement Learning

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***Abstract*—This project compares several policy parameter sharing strategies in a cooperative multi-agent reinforcement learning environment. Three policy structures are considered: a fully shared policy, a shared policy with a one-hot agent identity input, and independent policies for individual agents. The policies are trained using a Proximal Policy Optimization based actor-critic implementation in the MPE Simple Spread environment.**

The experiment contains three agents and three landmarks. Each agent observes all three landmarks but only one neighboring agent, so the interaction between agents remains partially observable. Each method was trained for 90,000 agent steps using one random seed. The shared policy achieved a best evaluation return of -63.17, while the shared policy with agent identity achieved -62.30. The independent policies achieved -70.78. All three methods improved over the random-policy baseline of approximately -98.39.

However, none of the trained policies achieved reliable task success. The agents learned to move closer to the landmarks, but they did not consistently cover all three landmarks at the same time. The results suggest that parameter sharing improves learning efficiency in this small homogeneous task, while successful coordination still requires better credit assignment or role assignment.

***Index Terms*—multi-agent reinforcement learning, parameter sharing, PPO, cooperative learning, partial observability**

I. Introduction

Multi-agent reinforcement learning (MARL) considers problems in which multiple agents learn while interacting with the same environment. Compared with single-agent reinforcement learning, MARL has additional difficulties because each agent changes its behavior during training. From the viewpoint of one agent, the changing policies of other agents make the environment non-stationary [**?**].

Policy parameter sharing is a common approach when the agents have the same observation space, action space, and objective. Instead of learning a separate neural network for each agent, all agents use one shared policy. This reduces the total number of trainable parameters and allows experiences collected by different agents to update the same network [**?**].

Parameter sharing may also limit the diversity of agent behavior. If two agents receive similar observations and use the same policy, they may select similar actions even when they should move toward different targets. One method for reducing this problem is agent indication, where an agent-specific one-hot vector is appended to the observation [**?**].

This project compares three policy structures:

1) *Shared*: all agents use the same actor and critic.
2) *Shared + Agent ID*: all agents use one actor and critic, but a one-hot agent identity is added to the observation.
3) *Independent*: every agent has a separate actor, critic, and optimizer.

The main questions are:

- Does parameter sharing improve evaluation return?
- Does agent identity help a shared policy learn different behaviors?
- How does policy sharing affect the number of trainable parameters?
- Does a higher distance-based reward lead to successful landmark coverage?

This is a small empirical project rather than a proposal of a new MARL algorithm. The main goal is to implement the three policy structures under the same PPO training conditions and compare their behavior.

II. Related Work

A. Multi-Agent Reinforcement Learning

Lowe et al. introduced Multi-Agent Deep Deterministic Policy Gradient and the Multi-Agent Particle Environment (MPE) [**?**]. Their work discussed several difficulties in MARL, including non-stationarity and coordination between multiple policies. The Simple Spread environment used in this project is a cooperative task based on MPE.

PettingZoo provides a common API for multi-agent reinforcement learning environments [**?**]. The current implementation uses MPE2, which follows the PettingZoo parallel API. At each environment step, actions are selected for all active agents and passed to the environment together.

B. Proximal Policy Optimization

Proximal Policy Optimization (PPO) is an on-policy policy-gradient method proposed by Schulman et al. [**?**]. PPO performs multiple minibatch updates using collected trajectories while limiting excessively large policy changes.

The clipped PPO objective is

$$L^{\mathrm{clip}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta)\hat{A}_t, \mathrm{clip}\left(r_t(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}_t \right) \right], \tag{1}$$

where $r_t(\theta)$ is the probability ratio between the new and previous policies, and $\hat{A}_t$ is an advantage estimate.

The current implementation uses a local actor and local critic. For the independent method, each agent updates a separate actor-critic network. For the shared methods, trajectories collected from all agents are combined to update one network.

C. Policy Parameter Sharing

Parameter sharing can improve scalability when agents are homogeneous. Chu and Ye studied parameter-sharing actor-critic methods and reported improvements in learning efficiency and memory usage [**?**].

Terry et al. examined a representational limitation of complete parameter sharing [**?**]. When different agents receive the same input, one shared policy cannot directly produce agent-specific behaviors. Agent indication addresses this problem by adding an identifier to the observation.

The Shared + Agent ID method in this project follows this simple idea. It is compared with both complete parameter sharing and complete policy independence.

III. Method

A. Environment

The experiments use the MPE2 `simple_spread_v3` environment. The environment contains three agents and three landmarks. The cooperative objective is to move the agents so that different landmarks are covered while avoiding unnecessary collisions.

Each agent chooses one of five discrete actions:

- no movement,
- move left,
- move right,
- move down,
- move up.

An episode has a maximum length of 50 environment steps. The environment reward is negative, so values closer to zero indicate better performance.

Each agent observes all three landmarks but only one neighboring agent. Therefore, landmark information is available, but the complete configuration of the other agents is not observed. The local reward ratio is set to 0.1 so that the shared cooperative reward has a larger influence than the collision penalty.

Fig. 1 shows an example state.

The curriculum option provided by the environment is enabled. Curriculum stage 0 is used during the first 30% of training, and stage 1 is used during the remaining 70%. The same curriculum schedule is used for all three methods.

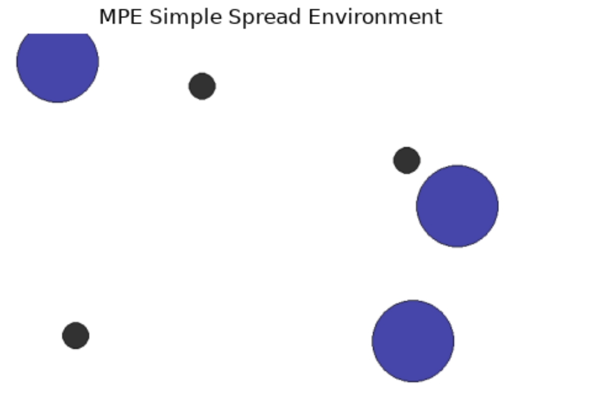

Fig. 1. MPE Simple Spread environment with three agents and three landmarks.

B. Policy Structures

The Shared method uses one actor-critic network for all agents. Experiences collected by all agents are combined into a single PPO batch.

The Shared + Agent ID method also uses one actor-critic network. A three-dimensional one-hot identity vector is appended to the observation. For example, the three identities are

$$[1, 0, 0], \quad [0, 1, 0], \quad [0, 0, 1]. \tag{2}$$

The Independent method uses three separate actor-critic networks. Each network is updated only from the trajectory of its corresponding agent.

The resulting trainable parameter counts are 140,806 for Shared, 142,342 for Shared + Agent ID, and 422,418 for Independent.

C. Actor-Critic Network

The actor and critic each contain two hidden layers with 256 units. The hyperbolic tangent activation function is used after each hidden layer.

The actor outputs five action logits. A categorical distribution is constructed from the logits and is used to sample discrete actions. The critic outputs one scalar value estimate.

The actor and critic use separate Adam optimizers. Orthogonal initialization, gradient clipping, PPO value clipping, entropy regularization, learning-rate decay, and target KL stopping are used to improve training stability.

Generalized Advantage Estimation (GAE) is used:

$$\hat{A}_t = \delta_t + \gamma\lambda(1 - d_t)\hat{A}_{t+1}, \tag{3}$$

where

$$\delta_t = r_t + \gamma V(o_{t+1})(1 - d_t) - V(o_t). \tag{4}$$

The discount factor is $\gamma = 0.99$ and the GAE parameter is $\lambda = 0.95$.

D. Reward Shaping

The original environment reward provides a dense signal based mainly on landmark coverage and collisions. However, the initial experiments improved the return without reaching the strict success condition. Additional shaping terms were therefore used during training.

For each environment state, the minimum one-to-one assignment distance between agents and landmarks is calculated. The assignment cost is

$$C_t = \min_{\sigma} \sum_{i=1}^{N} \left\| p_i^t - l_{\sigma(i)}^t \right\|_2 , \tag{5}$$

where p_i^t is the position of agent i, l_j^t is the position of landmark j, and σ is a permutation of the landmark indices.

Assignment progress is defined as

$$\Delta C_t = C_{t-1} - C_t. \tag{6}$$

A positive value means that the agents moved closer to a one-to-one landmark assignment.

The shaping reward is

$$r_t^{\text{shape}} = 3\Delta C_t + 2\Delta q_t + q_t + 20 I_t, \tag{7}$$

where q_t is the fraction of covered landmarks, Δq_t is its change, and I_t is one if all landmarks are covered. The final reward used for PPO is

$$\tilde{r}_{i,t} = 0.1 \left(r_{i,t}^{\text{env}} + r_t^{\text{shape}} \right). \tag{8}$$

The reported evaluation return uses only the original environment reward. The shaping reward is used only during training.

IV. Experimental Setup

The experiment uses one random seed for each method. This is sufficient for an initial course-project comparison, but it is not enough to make a strong statistical conclusion.

Each policy is trained for 90,000 agent steps. A rollout contains approximately 6,000 agent steps. Evaluation is performed every 15,000 agent steps using 10 fixed initial states. The checkpoint with the highest mean evaluation return is restored after training.

Table I summarizes the main settings.

The experiment was executed using a CUDA-enabled PyTorch environment. The complete comparison of the three methods required approximately 5.5 minutes on the current machine.

No external dataset was used. All samples were generated online through interaction with the MPE2 environment.

TABLE I
Main Experimental Settings

Parameter	Value
Agents / landmarks	3 / 3
Training seed	1
Agent steps per method	90,000
Rollout agent steps	6,000
Evaluation interval	15,000
Evaluation episodes	10
Maximum episode length	50
Observed agent neighbors	1
Observed landmarks	3
Hidden units per layer	256
Actor learning rate	3×10^{-4}
Critic learning rate	5×10^{-4}
Discount factor γ	0.99
GAE parameter λ	0.95
PPO clipping coefficient	0.2
Minibatch size	512
Update epochs	5
Reward scale	0.1

V. Results

A. Evaluation Return

Fig. 2 shows evaluation return during training. Each point is the mean of 10 evaluation episodes. Because only one training seed was used, the figure should be interpreted as an exploratory learning curve rather than an estimate of performance across random initializations.

All three trained methods improved over the random-policy baseline. The random policy obtained an average return of approximately -98.39.

The Shared policy obtained its best evaluation return at 48,000 agent steps. The Shared + Agent ID method continued improving until 90,000 agent steps. The Independent method obtained its best checkpoint at approximately 48,000 agent steps.

Table II summarizes the best checkpoints.

The standard deviation in Table II is calculated across the evaluation episodes, not across training seeds.

Fig. 3 compares the best returns.

B. Comparison of Policy Structures

The Shared + Agent ID method obtained the highest return, although its difference from the Shared method was small. Both shared methods performed better than the Independent method.

A likely explanation is that the shared policies receive experience from all three agents. All collected trajectories update one common network. In contrast, each independent policy receives only the data collected by one agent.

The Independent method used approximately three times more parameters but did not obtain a better result. Additional model capacity was therefore not enough to compensate for the reduced amount of experience available to each separate policy.

The agent identity input did not produce a large improvement. Its best return was only slightly higher than the Shared

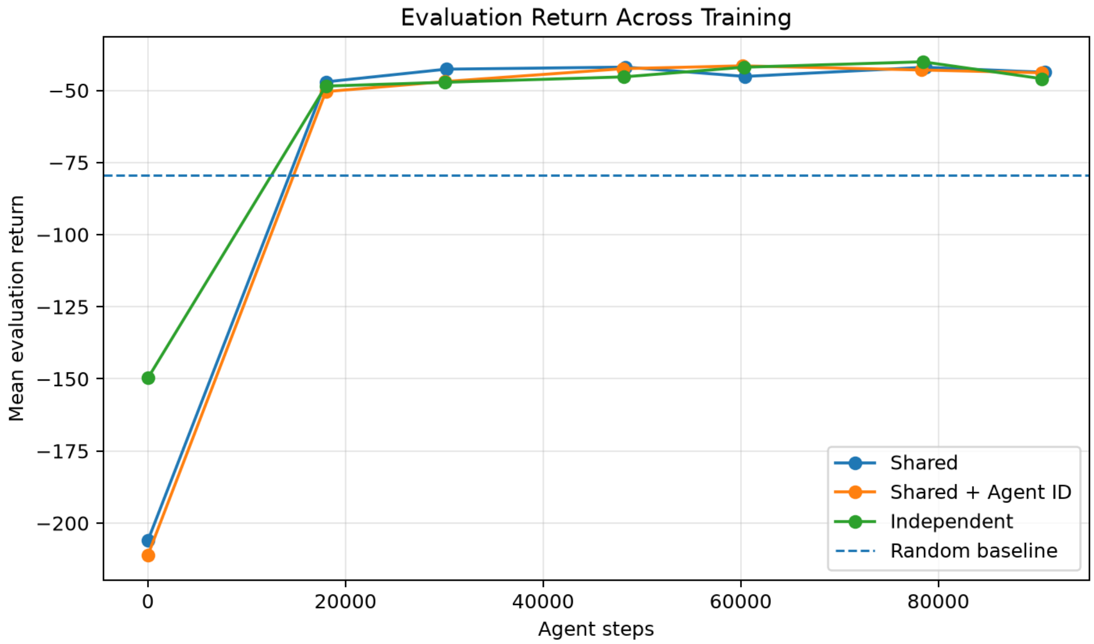

Fig. 2. Evaluation return during training for the three policy structures. Values closer to zero are better.

TABLE II
BEST-CHECKPOINT EVALUATION RESULTS FOR SEED 1

Method	Best Step	Evaluation Return	Coverage	Success	Action Diversity	Parameters
Shared	48,000	-63.17 ± 17.67	0.30	0.00	0.788	140,806
Shared + Agent ID	90,000	-62.30 ± 19.13	0.40	0.00	0.702	142,342
Independent	48,000	-70.78 ± 19.50	0.27	0.00	0.788	422,418

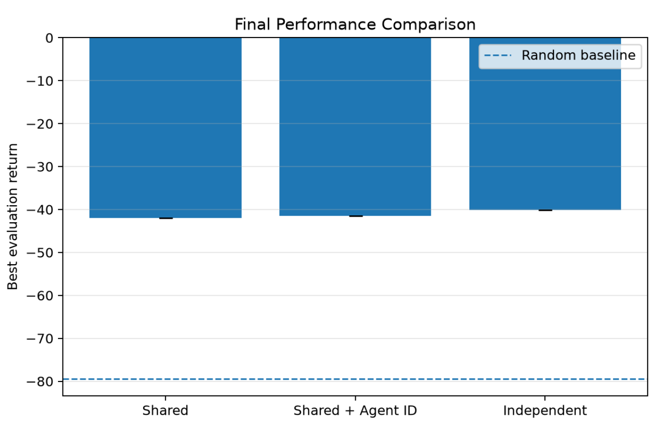

Fig. 3. Best evaluation return obtained by each policy structure.

result. This suggests that a fixed identity is less useful than information about the current target or role.

C. Coverage and Success

The success rate was zero for all three best checkpoints. All evaluation episodes reached the maximum length of 50 steps.

The coverage metric was higher than zero, which shows that the agents learned partial landmark coverage. Shared + Agent ID reached an average maximum coverage of approximately 0.40, while Shared reached 0.30 and Independent reached 0.27.

A coverage value of 0.40 does not mean that 40% of an episode was successful. It means that the maximum number of covered landmarks during an episode, divided by three, was approximately 0.40 on average. The policies therefore usually covered about one landmark and sometimes approached a second landmark.

The simulation videos showed that the agents moved more purposefully than the random policy. However, they did not maintain a stable one-to-one assignment between agents and landmarks.

VI. DISCUSSION

A. Effect of Parameter Sharing

The main result of the experiment is that both shared methods achieved better returns than the independent policies. They also used approximately one third of the parameters.

This result supports the idea that parameter sharing can improve sample efficiency when agents are homogeneous. A transition collected by any agent can contribute to the same shared network. For independent policies, the data are separated between three networks.

However, the current experiment used only one seed. The performance difference may change with another initialization or set of training episodes. The result should therefore be treated as preliminary.

B. Why Success Was Not Achieved

Increasing the episode horizon from 25 to 50 steps did not produce reliable success. Reward shaping also improved the return and coverage but did not solve the coordination problem.

One reason is that the shaping signal is shared by all agents. When one agent reduces the assignment cost, every agent receives the same additional reward. It remains difficult to determine which individual action was responsible for the improvement.

A second issue is that the minimum assignment is recalculated at every step. The implied assignment between agents and landmarks can change during an episode. The policy is therefore encouraged to reduce the global distance, but it is not given a fixed target landmark.

A third issue is partial observation of the other agents. Each agent observes only one neighboring agent. Even though all landmarks are visible, an agent cannot always determine which landmark is already being approached by an unobserved teammate.

The local critic also receives only the local observation. It may estimate distance-based improvement, but it has limited information about the complete cooperative state.

C. Reward and Success Metrics

The results show that evaluation return and task success measure different properties. The trained policies improved the dense environment reward from approximately -98.39 to about -62. Nevertheless, the binary success rate remained zero.

For this reason, only reporting return would give an incomplete interpretation. Coverage, episode length, success rate, and evaluation videos are needed to understand the learned behavior.

The current best checkpoint is selected using mean return. A checkpoint with slightly worse return but better landmark coverage would not necessarily be selected. A better selection rule could first prioritize success, then coverage, and finally return.

D. Limitations

The experiment has several limitations.

Only one environment and one random seed were used. The same PPO hyperparameters were applied to all policy structures. The reward shaping coefficients were chosen manually rather than through a systematic search.

The evaluation used the same fixed seed set during training. A separate set of test seeds should be used for final evaluation.

The curriculum schedule also introduces another experimental component. Although it is applied equally to every method, an experiment without curriculum would provide a simpler comparison.

Finally, the current feedforward actor cannot use observation history. This can be a limitation under partial observability.

VII. Potential Next Steps

The most direct next step is to use a fixed target assignment during each episode. At reset time, agents can be assigned to different landmarks using the minimum-distance assignment. Each agent can then receive an individual progress reward based on the distance to its assigned landmark.

This modification would provide clearer credit assignment:

$$r_{i,t}^{\text{target}} = \alpha \left(d_{i,t-1} - d_{i,t}\right) + \beta \mathbf{1}\left[d_{i,t} < \epsilon\right]. \tag{9}$$

A second extension is a centralized critic. The actor can still use local observations, while the critic receives the full positions of all agents and landmarks during training. This may improve value estimation and credit assignment without changing decentralized execution.

A recurrent actor using a GRU or LSTM could also use observation history to estimate the behavior of temporarily unobserved agents.

For a more reliable comparison, the experiment should be repeated with at least three random seeds. The final checkpoint should be evaluated on a separate set of unseen initial states.

Finally, an intermediate method between complete sharing and complete independence could be studied. Agents could share parameters or updates selectively according to trajectory similarity or estimated information usefulness.

VIII. Conclusion

This project compared Shared, Shared + Agent ID, and Independent actor-critic policies in the MPE Simple Spread environment.

The shared methods obtained better evaluation returns than the independent policies while using much fewer parameters. Shared + Agent ID achieved the best return of -62.30, and Shared achieved -63.17. The Independent method achieved -70.78. All trained methods outperformed the random baseline of approximately -98.39.

However, none of the methods achieved reliable success. The policies learned partial landmark coverage and improved movement behavior, but they did not consistently assign different agents to all three landmarks.

The experiment suggests that policy sharing is useful for sample efficiency in this small homogeneous setting. It also shows that dense reward improvement is not sufficient for solving the full cooperative task. Better role assignment and credit assignment are the most important directions for future work.

References